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Detection of *MCR-1* Gene in Multiple Drug Resistant *Escherichia coli* and *Klebsiella pneumoniae* in Human Clinical Samples from Peshawar, Pakistan

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Abstract: Background: The presence of plasmid mediated *mcr-1* gene in multidrug resistant Gram-negative bacteria poses a serious public health concern in today's world.

Objectives: The present study was aimed to detect the presence of plasmid mediated *mcr-1* encoding resistance to colistin in multiple drug resistant (MDR) *E. coli* and *K. pneumoniae* isolates.

Methods: A total 180 clinical isolates of *E. coli* (n=120) and *K. pneumoniae* (n=60) were isolated from different clinical specimens i.e. urine, blood, stool and pus, from diagnostic labs of two major public sector tertiary care hospitals in Peshawar, Pakistan. MDR profile of these isolates was assessed through Kirby-Baur disc diffusion method. All isolates were screened for colistin resistance by dilution methods. Colistin resistant isolates were subjected to PCR for *mcr-1* detection and confirmation was done by Sanger sequencing method.

Results: Overall 83.3% (100/120) *E. coli* and 93.3% (56/60) *K. pneumoniae* were detected as MDR. Colistin resistance was found in 23.3% (28/120) *E. coli* and 40% (24/60) *K. pneumoniae* isolates whereas *mcr-1* gene was detected in 10 out of 52 colistin resistant isolates including six *E. coli* and four *K. pneumoniae* isolates. Minimum inhibitory concentrations (MICs) of colistin in these ten *mcr-1* positive isolates ranged from 4µg/ml to 16µg/ml. All *mcr-1* positive isolates showed 99% sequence similarity when compared with other present sequences in GenBank.

Conclusion: Hence, our study confirms the presence of *mcr-1* mediated colistin resistance in the studied area. Therefore, urgently larger scale surveillance studies are recommended to investigate prevalence of *mcr-1* mediated colistin resistance and to prevent its further spread in the area.

Keywords: Multiple drug resistance, colistin resistance, *mcr-1*, *Escherichia coli*, *Klebsiella pneumoniae*.

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1. INTRODUCTION

Colistin, a polysaccharide antibiotic is revived in today's world as a last resort antibiotic due to the global dissemination of multiple drug resistant (MDR) Gram-negative bacteria [1]. Recently, the emergence of plasmid mediated colistin resistance gene *mcr-1* along with other *mcr* variants represents an alarming situation for human health [2]. The situation is becoming worrisome because colistin has been reintroduced after six decades for treating highly MDR Gram-negative bacteria. High resistance against colistin, reported from several countries has made clinicians worried about future scenarios where acute Gram-negative infections could not be treated effectively [3, 4]. The Gram-negative bacteria develop colistin resistance by using different mechanisms such as mutations in genes responsible for causing lipopolysaccharide biosynthesis or accumulation of acyl chain in lipid A portion of LPS layer [5]. The changes occurred due to the regulatory system of two components (PhoP/PhoQ and PmrA/PmrB), is another mechanism to develop resistance against colistin [6].

In past, it was believed that colistin resistance was only chromosomally mediated, but after detecting plasmid encoded *mcr-1* gene in *E. coli*, it has now been confirmed that colistin resistance is not just confined to chromosomal mutations but is also mediated by plasmid [7, 8]. After first report of *mcr-1* gene [7], nine other *mcr* variants have been identified, namely *mcr-2*, *mcr-3*, *mcr-4*, *mcr-5*, *mcr-6*, *mcr-7.1*, *mcr-8*, *mcr-9*, and *mcr-10* respectively with minor differences in their amino acid sequences [9-17].

Now as resistance to colistin is emerging worldwide, it is important to carry out in depth studies to find prevalence of these genes in different parts of the world including Pakistan. Therefore, the present study was designed to detect possible incidence of *mcr-1* gene in most prevalent *E. coli* and *K. pneumoniae* isolates from human clinical samples in Peshawar, Pakistan.

2. EXPERIMENTAL

2.1. MATERIALS & METHODS

2.1.1. Sample collection and identification

The present study was carried out from December 2017 to June 2018, a total of 180 non-duplicated clinical isolates of *E. coli* (n=120) and *K. pneumoniae* (n=60) were brought from clinical laboratories of two major public sector hospitals namely Hayatabad Medical Complex (HMC) and Khyber Teaching Hospital (KTH) in Peshawar, Pakistan. The isolates were found in clinical specimens of urine, blood, stool and pus swabs. All collected samples were immediately transported to Microbiology laboratory of Khyber Medical University Peshawar for further processing. The bacteria were isolated after culturing on Mac-Conkey agar, Cysteine Lactose Electrolyte Deficient agar, eosin methylene blue agar and blood agar medium. The isolates were identified by standard morphological and biochemical tests (API-10S system, bioMérieux, France) [4].

2.1.2. Antimicrobial susceptibility testing

Kirby-Bauer disc diffusion method was used to screen MDR *E. coli* and *K. pneumoniae* isolates against amoxicillin (AMC=30µg), ampicillin (AMP=10µg), amikacin (AMK=30µg), cefotaxime (CTX=30µg), ceftazidime (CAZ=30µg), ciprofloxacin (CIP=5µg), cefepime (FEP=30µg), colistin sulphate (CST=10µg), imipenem (IPM=10µg) and piperacillin/tazobactam (TZP=10µg) [18]. A standard 0.5 McFarland was made by mixing "0.05 mL of 1.175% barium chloride dehydrate with 9.95 mL of 1% sulfuric acid". Bacterial inoculum was prepared by touching four to five isolated colonies of the tested bacteria with the help of sterilized inoculating loop and transferred it to 2ml of sterilized normal saline. Suspension density of these inoculum was adjusted to 0.5 McFarland standard. Sterilised cotton swabs were used to spread the inoculums on plates containing Muller Hinton Agar (MHA). Antibiotic discs were placed using sterilised forceps according to the distance standardised by Clinical Laboratory Standards Institute (CLSI) guidelines [19]. MHA plates were incubated at 37°C for 24 hours. After that the zone of inhibition for different antibiotic discs were observed and measured according to CLSI guidelines.

Minimum Inhibitory Concentration (MIC) of colistin was assessed by using colistin sulphate powder (Sigma- Aldrich colistin sulphate), with both agar dilution and broth micro dilution method [20]. The European Committee of Antimicrobial Susceptibility Testing (EUCAST) guidelines were used to interpret observations and analyze results [21].

2.1.3. Plasmid DNA extraction

Plasmid DNA from colistin resistant isolates of *E. coli* and *K. pneumoniae* was extracted using alkaline lysis method [22]. To avoid contamination, the steps were repeated by following sterilized conditions to ensure the process accuracy.

2.1.4. Detection and confirmation of *mcr-1* gene

Polymerase Chain Reaction (PCR) was performed using *mcr-1* specific primers (Table 1), to screen colistin resistant isolates for possible presence of *mcr-1* gene [8]. PCR was independently performed and repeated three times under same thermo-cyclic conditions to confirm the reliability of the results.

Table 1. *mcr-1* specific primers used in PCR

Target gene	Primer Sequence (5' → 3')	Amplicon size	Reference
<i>mcr-1</i>	<i>mcr-1</i> F*: CGGTCAGTCCGTTTGTTTC <i>mcr-1</i> R†: CTTGGTCGGTCTGTAGGG	309bp	[7]

Footnote: * Forward, †Reverse.

2.1.5. Sequence analysis

The PCR products were subjected to Sanger sequencing to confirm presence of *mcr-1*. Sequencing results were analyzed by using the Finch TV software. These sequences were compared with the already reported sequences of *mcr-1* gene in GenBank using ncbi Basic Local Alignment Search Tool (<http://www.ncbi.nlm.nih.gov/blast/>).

2.1.6. Accession numbers

The sequence data from this study was submitted to ncbi GenBank under accession numbers (MK263178) and (MK340996) for *E. coli* & *K. pneumoniae* respectively.

3. RESULTS

In present study, a total of 180 clinical isolates of *E. coli* (n=120) and *K. pneumoniae* (n=60) were collected from two hospitals in Peshawar, Pakistan. From HMC hospital, 68 (56.6%) *E. coli* and 16 (26.6%) *K. pneumoniae* isolates, and from KTH, 52 (43.3%) *E. coli* and 44 (73.3%) *K. pneumoniae* isolates were collected. Among the total 120 *E. coli* isolates; 63% were isolated from urine, 3.3% from stool, 13.3% from pus and 20% from blood samples. Whereas, out of the total 60 *K. pneumoniae* isolates, 26.6% were isolated from urine, 13.3% from pus and 60% from blood samples.

Antimicrobial susceptibility test findings of *E. coli* (n=120) and *K. pneumoniae* (n=60) against ten antibiotics are given in (Table 2). *E. coli* showed highest resistivity against ampicillin (66.6%) and amoxicillin (56.6%) whereas lowest resistance against colistin sulphate (13.3%) and imipenem (20%). Whereas, *K. pneumoniae* showed highest resistance against amikacin (66.6%) and cefotaxime (60%) and lowest resistance against colistin sulphate (6.6%), imipenem (26.6%) and piperacillin/tazobactam (26.6%). Overall, 83.3% (100 out of 120) *E. coli* and 93.3% (56/60) *K. pneumoniae* showed resistance to at least three different types of antibiotic classes and were considered as MDR (Fig 1).

Table 2. Antimicrobial susceptibility patterns of *E. coli* and *K. pneumoniae*

Antibiotics	Antibiotic susceptibility profile of <i>E. coli</i> isolates		Antibiotic susceptibility profile of <i>K. pneumoniae</i> isolates	
	Sensitive	Resistant	Sensitive	Resistant
Amoxicillin	43.3% (52/120)	56.6% (68/120)	60.0% (36/60)	40.0% (24/60)
Amikacin	46.6% (56/120)	53.3% (64/120)	33.3% (20/60)	66.6% (40/60)
Ampicillin	33.3% (40/120)	66.6% (80/120)	46.6% (28/60)	53.3% (32/60)
Cefotaxime	53.3% (64/120)	46.6% (56/120)	40% (24/60)	60% (36/60)
Ceftazidime	63.3% (76/120)	36.6% (44/120)	46.6% (28/60)	53.3% (32/60)
Ciprofloxacin	46.6% (56/120)	53.3% (64/120)	46.6% (28/60)	53.3% (32/60)
Cefepime	66.6% (80/120)	33.3% (40/120)	53.3% (32/60)	46.6% (28/60)
Colistin sulphate	86.6% (104/120)	13.3% (16/120)	93.3% (56/60)	6.66% (4/60)
Imipenem	80.0% (96/120)	20.0% (24/120)	73.3% (44/60)	26.6% (16/60)
Piperacillin/tazobactam	60.0% (36/120)	40.0% (24/120)	73.3% (44/60)	26.6% (16/60)

*Interpretation of resistance in accordance with CLSI breakpoints.

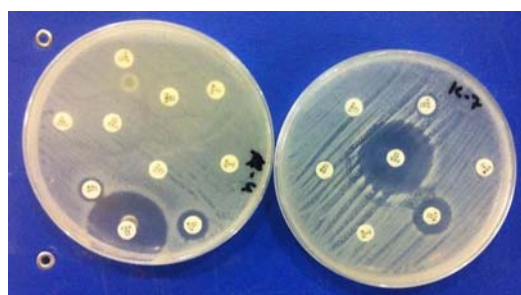


Fig 1. Antibiotic susceptibility testing (Kirby-Bauer disc diffusion method): highly resistant strain of *E. coli* and *K. pneumoniae* showing resistance to multiple antibiotics, reflected by growth inhibition zones around each antibiotic disc.

Overall, 23.3% (28 out of 120) *E. coli* isolates showed resistance against colistin with MIC values ranged from 4µg/ml to 16µg/ml whereas out of 60 *K. pneumoniae* isolates, 24 (40%) showed colistin resistance with MIC values ranged from 4µg/ml to 8µg/ml. The *mcr-1* gene was detected in 10 out of these 52 colistin resistant isolates including 21.4% (6/28) *E. coli* and 16.6% (4/24) *K. pneumoniae*. The details about *mcr-1* positive isolates are presented in Table 3. However, the remaining 42 colistin resistant isolates probably lacked *mcr-1* gene which indicated that resistance in these isolates could be due to chromosomal mutations or other variants of *mcr* family. PCR results of *mcr-1* positive isolates are presented in (Fig. 2).

Table 3. Characteristic and resistance profile of *mcr-1* positive *E. coli* and *K. pneumoniae* isolates.

Bacteria	<i>Mcr-1</i> positive isolates	Source	Resistance profile of <i>mcr-1</i> positive isolates to other antibiotics
<i>E. coli</i>	E4	Urine	AMP, AMC, AMK, CAZ, CTX, CIP, FEP, CST, IPM, TZP
	E5	Blood	AMP, AMC, AMK, CAZ, CTX, CIP, TZP, CST
	E6	Urine	AMP, AMC, AMK, CST, FEP, IPM, CIP
	E7	Urine	AMP, AMC, CAZ, CTX, FEP, TZP
	E11	Urine	AMP, AMK, CTX, FEP, CIP, TZP
	E14	Urine	AMC, AMK, CTX, CST, TZP, CST
<i>K. pneumoniae</i>	K4	Blood	AMC, AMK, CTX, CST, TZP, CIP
	K5	Urine	AMK, AMP, CST, CTX, TZP
	K6	Urine	AMP, AMK, CAZ, CTX, FEP, TZP, IPM
	K15	Pus	AMC, AMK, CAZ, CTX, CIP, IPM, TZP

Footnote: AMC= amoxicillin, AMP= ampicillin, AMK= amikacin, CTX= cefotaxime, CAZ= ceftazidime, CIP= ciprofloxacin, FEP= cefepime, CST= colistin sulphate, IPM= imipenem, and TZP= piperacillin/tazobactam.

The sequence analysis confirmed 99% sequence similarity with *mcr-1* gene sequences from ncbi GenBank. The sequence data obtained from this study was submitted to ncbi GenBank and accession numbers were assigned.

4. DISCUSSION

During recent years, antimicrobial resistance (AMR) has emerged as a major health care problem across the globe. Pakistan is among the countries having enormous burden of multiple drug resistant (MDR) and pan drug resistant microbes, one of the leading causes of mortality and morbidity in the country [23, 24]. The emergence of *mcr-1* gene, in particular, is a threat to

public health. This is of prime concern because *mcr-1* gene is located on plasmid therefore it can be efficiently transferred to other similar or distantly related bacterial species [25, 26].

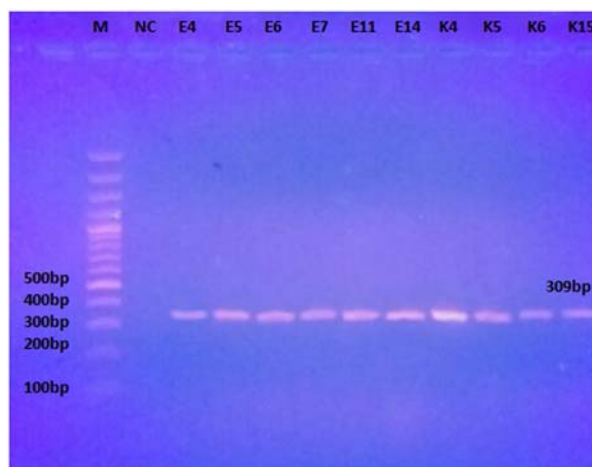


Fig 2. PCR based detection of *mcr-1* gene: Ethidium bromide stained 1% agarose gel showing amplified *mcr-1* fragments. From Left to right: M = 100bp DNA marker, NC = negative control, *E. coli* (E4, E5, E6, E7, E11, E14) and *K. pneumoniae* (K4, K5, K6, K15) isolates containing 309bp *mcr-1* fragment.

In present study, MDR detection was performed against ten important antibiotics including amoxicillin, amikacin, ampicillin, cefotaxime, ceftazidime, ciprofloxacin, cefepime, colistin sulphate, imipenem and tazobactam via disc diffusion method. Resistance against multiple antibiotics was observed in an alarmingly high percentage of studied isolates including, 83.3% *E. coli* and 93.3% *K. pneumoniae* isolates.

The detection of colistin resistance among the studied isolates was highly influenced by the choice of method used. Only two *E. coli* and one *K. pneumoniae* isolate showed colistin resistance through disc diffusion method. However, more isolates appeared as colistin resistant when tested by dilution method. The previously resistant isolates via disc diffusion method, were also appeared resistant when tested by dilution method. This observation is in accordance with a study conducted in 2016 [27]. The difference between the results of disc diffusion method and dilution methods in regard of colistin resistance lies probably because of the compromised sensitivity of disc diffusion method for the detection of colistin resistance. Therefore, dilution methods are consider as an indicator method for colistin susceptibility testing [28]. By agar dilution and broth micro dilution methods, 28 isolates of *E. coli* and 24 *K. pneumoniae* isolates were found as colistin resistant in our study. The similarity in the results of both these dilution methods in our study is in line with the findings proposed by Dafopoulou *et al.*, where both dilution methods showed same results [29].

The MIC values of colistin resistant *E. coli* in our study ranged from 4µg/ml to 16µg/ml, which were similar to the findings of two previous reports [30, 31]. However, these results were contrary to the MIC results of a study by Chan's research group [32]. In colistin resistant *K. pneumoniae* isolates, MIC ranged from 4µg/ml to 8µg/ml which is in line with a study report from Vietnam [33].

In the present study, out of 52 colistin resistant isolates, *mcr-1* was detected in 6 (21.4%) *E. coli* and 4 (16.6%) *K. pneumoniae* isolates. The high prevalence of *mcr-1* in *E. coli* as compared to other bacterial species was also reported by Wise *et al.* in a global surveillance programme [34]. The remaining 42 colistin resistant isolates in our study lacked *mcr-1* gene. The resistance in these isolates was probably due to other *mcr* variants or they might exhibit other mechanisms to develop resistance against colistin [35].

LIMITATIONS IN OUR STUDY

As the study was carried out under strict time frame and resources constrains, we were unable to perform the characterization of *mcr-1* positive isolates, detection of other *mcr* variants, and further gene-typing studies could not be initiated in the remaining 42 colistin resistant isolates.

CONCLUSION

Overall, the present study revealed the presence of *mcr-1* gene in MDR *E. coli* and *K. pneumoniae* isolated from Peshawar-Pakistan, which is a matter of serious concern because till-date colistin is used worldwide as a last treatment option against MDR Gram-negative bacteria. If careful efforts are not made to prevent misuse of colistin, we will be left with no treatment option to manage even simpler Gram-negative bacterial infections. Hence, it is concluded that urgent surveillance studies are

required on larger scale to find out colistin resistance in other regions of the country as well. It is very crucial to develop and implement preventive strategies against misuse and overuse of antibiotics in order to restrict the spread of resistance against different antibiotics, in particular, indiscriminate use of colistin in the farm animals and poultry must also be restricted.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

HUMAN AND ANIMAL RIGHTS

No animals/humans were used for studies that are the basis of this research.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

The data supporting the findings of this study are available within the article.

FUNDING

None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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